🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.1

Statistical Questions and Data

Lesson 8-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can tell the difference between a statistical question and a non-statistical question.

Language Objective: I can explain my reasoning using the words statistical question, data, variability, and survey.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Statistical Question | Data | Variability | Survey | Data distribution

👆 *Tap any word to see what it means and a picture.*

1. A question that expects a variety of answers and is answered by collecting data is a ________.

2. Facts or numbers collected to answer a question are ________.

3. How much the values in a data set differ from one another is the ________.

4. A way to collect data by asking many people the same question is a ________.

5. The way data values are spread out across a range is the ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

While sorting the questions, what clue tells you instantly that a question will need DATA from many players to answer?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Statistical Question** — A question where the answers will be different for different people.
  Pregunta estadística — Una pregunta cuyas respuestas serán distintas para distintas personas.
- [ ] **Data** — Facts and numbers you collect, like answers to a survey.
  Datos — Datos y números que recoges, como respuestas a una encuesta.
- [ ] **Variability** — How spread out the numbers are.
  Variabilidad — Qué tan separados están los números.
- [ ] **Survey** — Collecting facts by asking people questions.
  Encuesta — Recoger datos haciendo preguntas a las personas.
- [ ] **Data distribution** — How the data looks: where it sits and how spread out it is.
  Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

I know it needs data because it asks about ____.
Sé que necesita datos porque pregunta sobre ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A statistical question expects answers that vary across many cases.

Evidence: Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

Reasoning: This evidence connects the definition of statistical question to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I know it needs data because it asks about ____.**
  Sé que necesita datos porque pregunta sobre ____.
- **The answers will be different because ____.**
  Las respuestas serán diferentes porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Statistical Question
2. **Notes 2:** Data
3. **Notes 3:** Variability
4. **Notes 4:** Survey
5. **Notes 5:** Data distribution
6. **Watch for this mistake:** *A common mistake in Statistical Questions and Data is judging a question by its topic or its numbers instead of its variability. Students see a question about ONE player, ONE game, or ONE fixed fact — like "How many points does the team need to win a game?" — and call it statistical just because it has a number or is about sports. Or they see a question about a single player's whole season, like "How many touchdowns did the quarterback throw this season?", and call it NOT statistical because only one player is named, forgetting that a season covers many games with a different result each time. Both errors skip the real test: will the answers actually differ from case to case (game to game, player to player)? If yes, it is a statistical question; if there is one fixed answer, it is not.*
7. **Try It 1:** A. How many rebounds does each player grab per game? — *"How many rebounds does each player grab per game?" anticipates variability — different players grab different numbers of rebounds, so you must collect data. The other three each have one fixed answer.*
8. **Try It 2:** A. Because the warm-up times will vary from one athlete to another — *A statistical question anticipates variability. Different athletes warm up for different amounts of time, so the data you collect will vary.*